

Indian Institute of Technology, Gandhinagar
EE226 Semiconductor Devices
Semester 2

Note: Write down answers very briefly (1-2 lines explanations only)

From experimental Diode I-V characteristics:

1. Plot (linear and log scale) and compare the diode I-V with the ideal diode current $I = I_0 \left(e^{\frac{V_a}{\eta V_T}} - 1 \right)$. List down the regions (in terms of voltage ranges) where deviations occur from ideal current in forward and reverse bias.
2. Using the diode I-V in forward bias, extract the following parameters
 - a. Reverse saturation current
 - b. Ideality factor (η) –
 - i. You might observe different η values in different voltage ranges - Describe these ideal/non-ideal regions physical operation
 - c. Series resistance
 - d. Find the $\frac{dV}{d(\log_{10} I)}$ (mV/decade) in different regions of operations. How much is it in ideal (diffusion) limited current?
3. Extract and compare the built-in voltage with different methods listed below. Also describe the errors you might see in these different methods.
 - a. Plot linear I-V, extrapolate and find
 - b. Plot differential conductance (dI/dV) in forward bias and find
 - c. Plot $\ln(I)$ vs V extrapolate and find with and without series resistance correction
4. Find the following for the Silicon diode using reverse bias current.
 - a. Describe the effect on Reverse saturation current in reverse bias? Is it constant?
 - b. Depletion width vs Reverse Bias Voltage. Plot and observe.
 - c. Assume P and N doping is same. Plot and observe Doping vs Reverse bias voltage. You may Average out for all voltages to get a single value of doping (approximately).
 - d. What would be built-in voltage calculated with this method?
 - e. Calculate what would be the area of the diode?

Nanohub diode I-V Simulator:

1. Plot I-V characteristics for Silicon in (linear, log) for different doping and compare – which regions are affected the most? Why?
2. Plot I-V characteristics for Silicon, Germanium and GaAs. Plot and compare which regions are affected the most? Why?
3. Tune and observe the effect of the R-G lifetimes on Si diode I-V characteristics. Which region of the I-V is affected most?
4. Increase the temperature and plot the change in I-V.
 - a. List out the reasons for this increase/decrease in current in FB and RB.
 - b. Find the $\frac{dV}{d(\log_{10} I)}$ (mV/decade) with temperature from the plot
5. Describe a method to find out built-in voltage using temperature dependent I-V. Calculate.
6. Plot C-V. Describe a method to find out built-in voltage using C-V. Compare with (5) and any of other methods.

Comparing Experimental diode, I-V with Nanohub Simulator diode I-V characteristics. Use appropriate assumptions

Emulate, match and compare the diode Forward bias experimental I-V with the Nanohub diode I-V simulator.

1. Tune and observe and list which parameters deviate significantly and calibrated to get a closer match with experimental I-V
2. As simulation provides current density, you would need to multiply the same with area to get the final current and compare with experimental iv. What would be the approximate area of the diode?